

ActInf Textbook Group ~ Cohort 2 ~ Meeting 22 (Chapter 10, part 1)

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all right greetings thanks for joining all it's march 29th 2023 we're in cohort two
in our first discussion on chapter 10. so let's head over to chapter 10 and um begin with any
comments really about anything within the active area um several were voiced before we begun so
perhaps terry feel free to begin with um what you wanted to point towards and let's just see where
that goes thank you uh in essence my my question is about the concept of a um a system of volitional
consciousness behind that sits behind the uh nodal active inference model and creating
um uh i suppose uh making decisions about about the strategic direction of the organism and um i find
just repeatedly within this chapter there seems to be this reference to you you know a a conscious
entity behind the algorithm and my sense from trying to understand this material is that it's almost
like um in meditation where you know the the the the experience is the experience that that that almost
there is no eye behind the experience and that um it just jars with me when we're presented with a
almost complete uh active inference model or that that where where our our actions in the world are
actually experiments to gain information about the external state to um up ultimately update the
model and what consciousness is as you said at the start is is a difficult concept but um it seems
strange to me within the context of this model that we refer back to some sort of uh underpinning
strategic conscious experiential driver of action when that seems out with this model

uh

or unnecessary for this model

hmm

uh michael or ali want to give any first thoughts on this

yeah ali go for it

uh

uh

yeah uh well uh actually i don't think uh they necessarily uh somehow uh i mean the active inference
framework uh doesn't require any conscious entity in the sense uh of our anthropic sense of this word
uh consciousness because uh what fp actually tries to model is a kind of uh quote unquote as if epistemology
so by uh as if epistemology it doesn't mean that uh everything that uh can can be modeled with uh fp
uh from uh from uh from uh from uh from uh from uh from uh from uh from uh from uh from uh from uh
from uh from uh from uh from uh from uh from uh from uh from uh from uh from uh from uh from uh from uh
modeled so i'm sure you've encountered this word self evidencing throughout
this book which refers to the same concept of as if epistemology so it's or
one of i'm sure had Daniel's favorite figures in the book figure yeah 9.1 also captures this sense
of perspectival view view of kind of the nature of this modeling is not that we have this sense of
omniscience that that can be applied to the environment or that can be
reasoned through with i mean reason through with and that can also be applied to our sensorium or

perceptorium but it's like we have this model inside a kind of apparatus that can see its behavior as if following through a kind of trajectory in the probability density space so by volition that you're referring to it's my understanding that it's also another term that they've also used that's related to this self evidencing is they is the object or the agent tries to infer its environment through a kind of what was this term it wasn't something like indirect person no it wasn't um Daniel can you help me find the term for uh the indirect perception of the specific term that's used in the literature to refer to indirect perception uh sorry just a second I don't exactly remember okay I'll find it and I'll get back to you in just one second

okay and I mean I I I I stop me if I'm going down some unnecessary rabbit hole um uh but it's almost as if uh this um

um system uh active inference represents a model by which one could uh uh create a psychological zombie um in in in that you know by active inference without any um sense of internal self one could behave in the world uh by um you know minimizing um uh variational free energy uh in the way that we behave in the world and um

Daniel Dennett seemed to in one of his books sort of talk about that um uh the are are are you know that we have active won't rather than active um uh that than rather than free will you know and um i i i suppose the the the implication of that is that we um that that our conscious reality is is like a a story we tell ourselves almost uh shortly after the event and yet in this book there is this reference to uh uh uh a conscious will within the within the framework or certainly within that last chapter uh chapter um five that we're isn't it chapter five we're on today so i i i don't i i i i mean i don't have this worked out of my head but um i'm just interested it's it's entirely possible that um one could say active inference isn't about what it it it doesn't speak to consciousness consciousness is another thing active inference just off gives another is a potential model that uh drives and decision making within the organism almost you you you uh i i i i and that's okay um but i'm just in i i'm i'm just interested in in other people's thoughts on this it's it's it's great topics so just a few notes um

broadly there are perspectives on consciousness well first i'll just note consciousness appears to be only used once in the book so while it is true that um it's an area of interest and was the live stream with uh vanja visa this morning i mean and the authors all explore elsewhere but any interpretation of consciousness i i i don't even think it's within the it's not a pillar of the framework it's something that is like inferred towards so i think that will always leave a fundamental um uncertainty and pluralism and then we can think about at least a gradient of um accounts of consciousness in some accounts consciousness is um after the fact in some way that it could still be anticipatory from a predictive processing sense but it it does not have a causal power it's it's the a narrative told to um let's say minimize variational free energy to make sense of memories and so that kind of narrative expositional self as epiphenomena and experience without causal impact and then that is of course contrasted with causal accounts of consciousness and that's where whether you come from some philosophical tradition or certain folk psychological traditions um which both can arise during scientific writing as well things like emotions or awareness or even attention are actually

described as causally upstream of other phenomena and that puts one of course squarely in the kind of cartesian dualism area of philosophy like how does something that is not physical influence something that is physical um so the the the the dominant mode to me on that question is consciousness is something we can model about and around with active inference but no perspective on consciousness seems fundamental to using statistics in this way and so i expect and already find confirmatory evidence that some people take a more epiphenomenal approach to consciousness and some people taking a more causal approach

for example treating it like a hidden latent factor that does have a causal impact in a causal bayesian sense and then also just recognizing yeah we don't know exactly what it is but that's how we do active modeling we propose latent causes

and those become our generative model

so in some ways there may be more similarities between these two than their hyperbolic forms suggest and in

either case i feel like it's a continuum that's accommodated methodologically by what ali referred to as the as if epistemology and that i think connects to daniel dennett's intentional stance like i'm going to play this chess computer like it's trying to beat me if it's not it's then i then i want to win

so if it wants me to win too then that's going to be great but i'm going to hedge my bets and play like it's trying to win and i think that kind of um intentional stance is not just warranted it's actually supported by these methodological tools

terry or michael

well in in some ways and we're getting way off topic and then i will be quiet oh no this is

we're technically in chapter 10 so we're we're totally on just recap and meme status

okay but we but it's like but you would say that wouldn't you you know i i if consciousness isn't after the fact um uh story we tell ourselves um you know it it's at the model if the world is probably more likely that one finds oneself in front of a computer play playing chess uh because of the you know all of the previous events that have occurred during your your life and that you are now playing chess so the story you tell yourself is that i'm going to play to win but actually that it wasn't the it wasn't the conscious decision that got you there if consciousness is an after the fact uh uh

you know a hallucination or our story that we that we create to uh justify the things that are happening in the external in the unknown external state you know because the computer is in the unknown external state it's it's a it's something that we have internally created um uh as our model based on

the information we're getting across our markov blanket am i getting too mad no i'll give a short

note and then ali so um one answer from a biological perspective is especially over evolutionary time but also within a generation you will only repeatedly observe things that are fighting to live and

persisting amidst challenging and adversarial settings so if we just say oh well just the person in front of the computer what what would be subconsciously leading them to want to win but if we say we're in a chess tournament series and only people who have won previous chess tournaments defined by winning in this arbitrary way only they are persisting in this tournament well then over time you will come to find

that those agents are acting as if they want to win at chess now then there's the debate well what if they're just acting like they want to win at chess but they don't really want to win they're they're chess zombies they're just acting like it and then that's the pluralism some people look at that chess tournament sequence and they go yeah i actually have like reasons to think that people don't have experience or intentions or that consciousness is just a narrative told after the fact and so they're just acting like they want to win and the other person tells this extremely intimate causal story about how the the motivation of the person drives their action selection but in the end they're unified by this as if epistemology which is to say the intentional stance ali

uh yeah also this uh kind of as if epistemology um comes from uh at least partially comes from uh kantian transcendental idealism so for kant uh concepts do not uh apply to reality in the form in which uh we can only think them but um it is also urgently required for any theoretical practice to think and investigate um objects uh with the help of the concepts as if uh they were given or determined um in a specific way so uh this kind of as if epistemology um is you see uh concepts have to um acknowledge that reality is presupposed independently of the choice of its semantics or its meanings on the other hand uh in their self-reference and self-similarity uh they work with this heuristic of as if uh so that uh it can be constructed in this way as if reality and concept would actually coincide with each other but uh some recent philosophers such as uh karen broad attempts to overcome this kind of as if epistemology

uh for example in her agent realism agential realism um by means of construction of interactive uh discursive in which uh both epistemology and ontology can be seen as the very same thing and uh we wouldn't be required to draw a boundary between the perception on the one hand and the thing that we're trying to percept or model on the other hand so through this uh intra-action not interaction but intra-action or constructive interaction they constitute uh each other so as they give rise to their very existence and uh also i think in some view fp uh is perfectly compatible with this more um loose sense of uh onto epistemology because uh one of the requirements of the fp is if things exist how they should uh behave right so it somehow presupposes this kind of interactive uh this discourse uh by uh basically eliminating the boundary between the uh thing that we're trying to model and um i mean the the model the model or the map so that's basically uh the argument behind that map territory fallacy fallacy uh so uh yeah this is a bit uh more complicated uh in in this sense that uh if we want to get to the bottom of this kind of epistemology uh we will hit a dead end if we just um if we just uh go through this more traditional uh epistemology of uh of a western epistemology of uh distinguishing between the thing that's what we're trying to perceive and the perception itself uh but as i said fvp somehow tries to uh overcome this um problem in the first place um that's that's that's that's uh beautifully um uh uh described um and nicely closed down um and the thing that i was thinking initially was um the assumption that in the external state there's actually a chess competition happening um is assuming almost this uh car uh you know uh uh that that that our um internal state representation is an accurate or a true likeness of what the external state is which i think ali was kind of saying but um i my my sense is that it clearly can't be because all of our um all of our all of our qualia are um uh endeavors that the the the the

central nervous system uh makes to interpret um signals that are generated without the qualia that we apply to them

yeah no these are these are great one short thought then michael um yeah the internal state does not in the model does not have to have the same structure as the external state so like often we'll talk about well the external state in the model could be a continuous temperature and the internal state could just be maybe too hot just right or or too cold so they can have different structure and that reflects that they don't even have to be of the same type even within the model let alone the territory however by the cybernetic principles like good regulator theorem again eventually you can always just create an edge case where the agent has no um mapping between their internal cognitive dynamics and the external regularities like it's just playing the game randomly with respect to the environmental variability and in a research or an education setting that might be pedagogical but when we're talking about real measurements that we made even of like the the um pen on my desk it is resisting dissipation over some spatial temporal time scale and that is actually the surprising implication of if things exist how they should behave is like well i am repeatedly measuring this pen so now what can i say about that and it turns out you can actually say a lot if you bake into your understanding that you are observing something repeatedly existing and that's exactly what we see in figure 9.1

and so we we have a a unity of the scenario in our model

via articulating the partitions on a base graph

so that means there's still totally up in the air map and territory and all of that

from our HIM HIM HIM HIM HIM HIM HIM HIM HIM HIM HIM HIM HIM HIM HIM HIM

the these questions maybe somehow relate to this tale of two density papers um great you mentioned last time so and i my understanding of this paper is that they um try to argue that there is no representation in the mind but the mind is a controlled system and then they talk about two different distributions that i so so so one distribution is the recognition distribution and the other one is the generative system if i understood correctly but i i didn't really understand how this distribution how this this distribution comes into being through the um generative system so they equal one with

and enacted so the recognition system is the an embodied system and the generative system is the enacted system and my understanding is that this is all that there is so for instance they say here the on page 235 the generative model manifests as a control system that uses exploitable structural similarities encoded in the internal states of the organism so instead of having representations they talk about control states and

uh do you understand this correct so how can that be um a distribution uh or um i i didn't really i didn't really understand understand where these two distributions are great all great questions so i'll try to address some so

um

this question of exploitable similarity is very much like the good regulator theorem in cybernetics so it doesn't mean that the temperature in the room let's just say that it's physiologically relevant so there have to be some sensors and some processing apparatus and also action apparatus but at the very least just for cognition a sensory and processing apparatus that is able to like measure temperature it can be a proxy of it doesn't need to be strict isomorphism they say it's weaker than a strict isomorphism but it actually has to have some sort of causal relevance now if you're like well i'm in a simulation where the temperature is fluctuating but it has no survival value then it's like okay that's like saying you know cosmic rays are going through us they don't have survival value and so they're not part of the cognitive model in this setting um but um this points to this idea that there is information in the sensory apparatus about action it's actionable intelligence with respect to persistence why persistence well we could say in pursuit of reward but then we would need to ask how systems persist why persistence because we're at least repeatedly measuring the system so again with the as if epistemology it's as if in order to persist amidst dissipative and adversarial settings it's as if there's an exploitable information content in the perceptual Umwelt that enables that kind of thing to persist that's like the tautology that also goes beyond a tautology now to the um specifics of the the tale of two densities um this also might have been the first dot zero in in six i'm not sure in bayesian statistical analysis

models can be run in two different directions this could be like a hyper prior and and um here's the hyper priors on the right and the data are on the left so it's like you can specify the hyper prior or the hidden state distribution and you can emit pseudodata that's called generative or you can have data and then parameterize hidden states that's the recognition model so this could be the actual measurements of children's height in the classroom here's our generative statistical model gaussian family mean and variance so given a mean and a variance estimator two numbers metaphysical statistical we can generate infinite class size of synthetic height data or from one height observation or more we can update our generative model of the world okay then more broadly we're interested in this relationship between the world and the agent agent in the niche so perception and how that updates the agent's the agent's learning and uh and perception and then how the agent acts on the world and what they argue in this paper

is that the recognition model in a bayesian statistical setting so going from measurements which might be noisy and the the height in centimeters isn't like the real height of the person the number on the scale

isn't the weight the number on the thermometer is not the temperature like it's data this is why the empirical perspective on behavioral modeling is so essential because when we go on this bayesian side it's really easy to talk about how data are recognized and update hyper parameters and then hyper parameters can be in the business of continually generating unfolding predictions

we have a noisy camera with a blind spot we're updating our generative model of vision and then we're going to be in this case just creating more synthetic vision data

in an activism and all of the associated ease embodiment etc etc etc etc they are focused on

on the materiality of the engagement of the agent with the world in this paper tale of two densities
they draw these two together to argue in the context of what we'll just point to in figure 4.3
the a matrix here
with the a matrix here two
there's a generative production of synthetic observations you could use it that way that's
the generative direction of a or you could use a in its recognition capacity to update hidden states from
observations so even without getting into the generative action which is what in the inactivists
setting what they're most interested in is the actual causal consequences of action just within a figure
um 4.3 hidden state inference we can think about a tale of two densities
and about how bayesian statistics helps reconcile this tale of two densities
because the same generative model which is just what we call whatever bayes graph you construct the
generative model
can be used equally flexibly to do inference about any part of it like maybe um you know the hidden states
but you're trying to determine the prior so these graphs are agnostic as to what you know and what you want
to infer
now it turns out that for real settings that we find in the behavioral lab and out in the fields
we get observations
we don't have access to hidden states but we want to infer them
we don't even think a or b matrices or d matrices exist in the world we purely want to infer them
as statistical parameters and then we want to know about action
and in some settings we have the um organisms trace of actions and we're trying to infer their cognitive
parameters that's the figure 9.1 setting we saw their actions it was data for us and we're looking to
parameterize a cognitive model for them another setting is we don't know what action will be taken
because we're the ones taking it and that is like a prospective application of these types of models
to take ongoing streams of observations and then perform action selection in the moment
but these frameworks are enormously general with whether they're used for online or offline inference
and like what you actually have access to measuring and how
so little long but the tale of two densities is about how the bayesian epistemology
and structuring of the relationship between observations and inferences about observations
maps in the inactivist setting
and they're not comparing to some aspects of how we think about perception cognition and action in the
niche
and their paper
positions that
and they're not comparing to that within the debates around representation
because representation are although there's different perspectives etc postulated to be like
the on agent
on agent

aspects

that have semantics about the world

yeah

and they say

the agent just controls

the world through their own actions

that's my understanding

yeah

may be stretching the tale of two densities, making more of a set of novels around densities because it's just ultimately a cultural reference. The technical point is that these variables are statistical distributions and hence they are densities. They're statistical densities over some support. So the two densities that we might be interested in or amongst two of the ones that we're interested in making a unified account of are the perceptual density, which is about how data map on to hidden states. Like given retinal data, is that a car? That's a recognition model. And then there's kind of like two kinds of generative densities. You could run that car recognition model backwards. So given the label of a car, generate me cars, stable diffusion, generative AI. But then there's also this action as a generative step that brings in the actionable and cybernetic constraints, which is why they focus on the exploitable similarity. But in general, Bayesian models, you can do this kind of observable hidden state inference about anything. The organism, Bayesian brain, or if we choose to use Bayesian statistics, it's going to be doing something like that. The world has a real temperature, which is like molecules moving at different speeds. Perception occurs through sensors, which transform whatever it is the molecules are doing into something totally, totally different. Can I just make a little point about that? That I remember watching, you know, Richard Feynman, just a short video, where he was describing temperature is actually infrared radiation, which is part of the electromagnetic spectrum. So on one part of that spectrum, we see light, and on the other, we experience temperature. So temperature is actually a proxy for something else. You know, the mechanism that we have to perceive temperature is then a proxy for the amount of energy within that, for example, room. So it's not, our response to the infrared information is a response that's not based on infrared, necessarily that infrared is going to damage us. It's that there's too much energy in this room. And the tool that I have to assess that energy in the room is actually monitoring the infrared radiation, which I can't see, but I can feel. Any comment on that? Yeah. Like by analogy, how about tasting different foods? And there might be a toxin, like we can't directly tell that something is going to be contaminated, but over time, successful entities in an unchanging niche. Now that's not every case, but just with that simple case at first, successful entities in an unchanging niche will come to exploit similarities, which is to say mutual informations between the recognition model and the actionability. And so they don't need to be directly measuring or even internally representing, this is going to make me sick, or this is the wavelength of light. Any and all informative cues of sensors are going to be aligned towards adaptive action. That's the synthesis of signal processing and control theory in active inference. And it turns out that by taking a first principled approach

to the synthesis of those two fields, again, just coarsely, we have new insights on the question of representation

in cognitive science.

Yes. But taste is a very interesting phenomenon in that we have sweet, salt, sour, bitter, and yet there are, I think there are trillions of different tastes that we can actually create or,

or I think smell is probably associated with taste as well. So, you know, from a limited amount of sensory information, we can create a vast array of experience.

Yeah. I mean, with the, um, the Jolly Ranchers, I don't know, just a candy I had in childhood. Like they literally taste the same if you don't see the color, but if you do see the color, even if you try to forget, they will taste different. So yeah. So different, um, that's like referred to a sensor fusion, different sensors come together in the generative model.

Yeah. And it's, it really speaks to the prediction.

Yeah.

You know, the prediction model, it really speaks to what are our, what are our, uh, priors, you know?

Another kind of mind bender on this is like the visual example, there's so much abundant, um, on ramps to predictive processing. Like we don't notice our blind spot. We don't have any blur when we do the ice cates several times per second because of sensory attenuation, you know, color vision and precision in the periphery. And okay. I'm on board with a generative model of vision.

I'm I've kind of dispelled this sort of incoming information camera processing model of vision.

I really believe in the visual generative model getting updated. And it's like, but taste it's like, yeah, by extension, what does that really mean that we're generating our taste experience?

Well, then why wouldn't I just generate tastes that tastes amazing? Like, well, why, what, why wouldn't your generative model of vision just generate what you like to see?

And that's where the real organismal constraints and where appeals to our development, ecology, and evolution do come into play. So again, contrived Gdenkin can be proposed absent these super obvious constraints for real systems. However, in the context of empirical behavioral modeling, it's usually sufficient to say the prior distribution on this or the trade-off between these two different drives is inherited from history and development or even from intergenerational processes.

So, I think that's the case.

But, but, but isn't, isn't it that we generate a good taste going to a baker where we like to go and doing and going towards those areas where we get those kinds of rewards that we prefer?

I mean, doesn't it make sense also in that?

Always coming back to the baker.

I don't know why.

Yes. Okay. Okay. Here we were. Okay. Chapter seven. We were, we were engaged in a policy vacillation. We were, we were not sure if we were wanting to get bread or if we want to go to the barber. So now, the question of the taste. Yes. If we expect and prefer a given taste like sweet, and I think this

is, holds true for sugar, but it can hold true for other things too. Then in the free energy evaluation of different policies with expected free energy, expecting observations that align with your preferences will be associated with pragmatic value. So that is how preference for sweeter tastes puts its thumb on the scale of policy selection, biases policy selection towards tasting sugar as a pragmatic reward.

So that happens amidst a relationship with epistemic value as well.

So it's like, um, epistemic value could be pursued.

Like we need to find out how to get to the baker.

I mean, there's, there's many ways to go, but yes, you would parameterize a generative model with a preference. Let's think about this, this case with sugar and the baker in those two settings that we described. So in one case we're in figure 9.1 and we're, we're observing people's behavior. So we have their credit card history. Whoops. And we're observing who is going to the baker or what they're ordering or what they're clicking on online. And now we're going to do inference on a generative model for them. It's the, as if generative model. And it's like, wow, this person has a very strong C vector, very strong preference vector for sugar. Like given alternatives, they're always going for the sweeter one. And this person has a neutral preference for sugar. And this person actually like moves away from sugar as represented by our parameterization of the C vector.

So that's one setting. We have data on a bunch of different entities on what they chose in different sugar concentrations.

And then the other setting would be now step into the trajectory of one of those cognitive models. And now ask what your next action should be.

And that's the, as if a gentle or intentional stance. So this is the behavioral scientists view from the outside. And then that is going to be flexible with allowing us also to take views as if from the inside.

And just because we can take an as if view from the inside

the outside doesn't mean that that's the territory. That's literally the map territory fallacy.

But it means that within the limits of our cognitive model, we do have that opportunity.

One of the things that strikes me as you talk about this is that

one of the interviews Carl Friston has done online, he talks about, you know, our actions during during the day that, you know, we get up and we have our breakfast and then we, you know, we go to the bus and we get go to work and we do our work. But we, you know, and actually, most of our life is little routines, we think that we have these independent lives where we are constantly making decisions about what, you know, what, what do I want to eat today? What, where will I go? But actually, the person who ends up going to the baker will find that on their walk to the bus stop, they pass the bakery, and they get a trigger from the smell of the bakery to do the thing that, you know, to predict that it's just interesting that we don't exist as these completely independent, no matter where you live, no matter who you are, there are so many aspects of your life that are just so so repeatable that we actually live within a very constricted box, apart from odd times when we step outside that box. I don't know if that's relevant to what that picture, but

it's awesome. I mean, very interesting to point to the state of exception. In livestream 10, with Mao and Ramstad et al, who are the other authors? Mao, Axel, Karl and Maxwell. They talk about like strong and weak scripts. A strong script in a social setting would be like if we were literally reading from a screenplay, or if we were doing some sort of highly ritualized behavior, whereas the weaker script concept refers to like norms, normativity, expectations, x kind of person should act like this, I'm x kind of person, I would be unsurprised by doing x, oh, what do you know? I'm doing x. And that kind of progressive identity capture and development of understanding what kind of thing one is. And then what are the two ways you can reduce your discrepancy? You can change, you can learn, change what you think that kind of thing is, and you can act. So undergrads should act this way. You know, I expect, here's the distribution of undergrads. If one's understanding of what an undergrad was, was totally flat with respect to everything, there's nothing that characterizes an undergrad versus anything else, just from one person's perspective. Well then that aspect of their identity wouldn't shape their policy selection. Whereas if someone says, well, as a x, I should be doing y. That can be approached from a purely statistical deflationary perspective. Like, I just expect myself to be doing this. Like, or it can go even beyond that too. So these are all very relevant. And learning about the fundamentals of this kind of statistical modeling help us, I hope, I prefer also, that it helps us actually respect the complexity and the richness of these phenomena, which just simply cannot be expressed in words or math, which is why we talk about the map.

Yeah. One of my wonderings about all this, because I do find the maths and the the pictures and the diagrams all very difficult to grasp. In some way, it is there a potential that if we really understood the model, we could start to use that as a system to inform our understanding of individual behaviors. I mean, I'm a pain doctor. So I am the my, my, my interest in all this is this, this, this, this truth that we can alter our, you know, if you're in pain, if you have back pain, you can alter your prediction about your back pain and actually alter your back pain. And I'm just wondering if this kind of modeling could in some way give us a tool to help us to support other humans in understanding their ability that they can update their priors, you know? Yeah, it's a great vision. I think the closest that we've explored that is with the help of Ian Tennant, who has a interoception focus, and this therapeutic alliance concept where, where a lot of these kind of avenues and like, it's not as simple as just wish it away any more than the generative model of vision would be like, I want to see, you know, a unicorn in front of me. So how do we reconcile that agential realism that Ali mentioned earlier, like this active co-construction and negotiation and sense-making from our sensorium and the ways that our narratives and the scripts and the expectations shape our inference and action and choose to have a preference and to act to push things in a certain way? Yeah. Absolutely. Yeah.

But I suppose what I'm kind of thinking is, could you have a, like a, you know, a Sam Altman created model, where you run the model and that can inform how, that can inform the, uh, understanding of an individual's, how they got there and potentially offer some insights into what to do about it? You know, that,

that could this speak to a model that could be run where we could kind of do the experiment and then that supports an individual on actually going on the course of action to change their experience. That, that's, that's kind of where I, what, what, what really blows my mind about this and that in the current world that has only come into existence in the past month or so, this actually seems almost plausible because of the, the similar layered nodal systems that the likes of ChatTP are using to protect language.

Yes. It's a great.

It's not active inference. You know, it's not active inference. You could model it around a notion of active and the training of the system.

There's so many avenues to explore there.

In our discussion next week,

let's look at that again.

But absolutely, with respect to figure 9.1s about us, the data exists, the means, motive, opportunity, anyone who wants to.

Can do things.

And that's a big question.

So for next week.

Thank you, though.

I'll stop recording.